ESP32 Li-Ion Battery Power Management Module

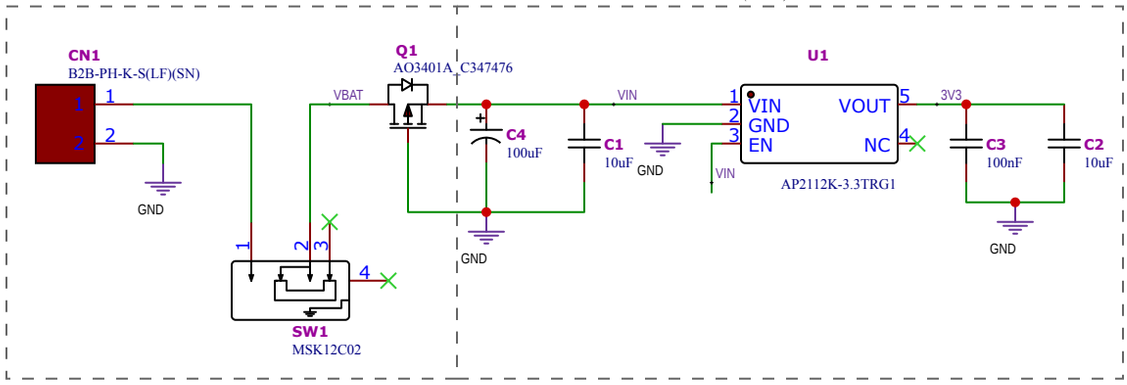
Features:

- *USB-C Li-Ion charging (TP4056, 580mA)*
- *Reverse polarity protection (P-MOSFET, 22mV drop)*
- *3.3V LDO regulation (AP2112K, 600mA, 70dB PSRR)*
- *Battery voltage monitoring (resistive divider to ADC)*
- *Power ON/OFF switch*
- *Power status LED + Charge status LEDs*

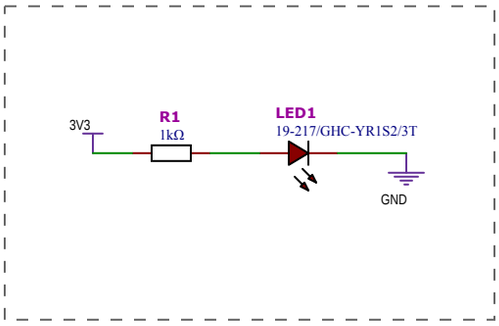
Input: 1S Li-Ion 3.0V–4.2V │ Output: 3.3V @ 600mA

1.BATTERY INPUT & PROTECTION

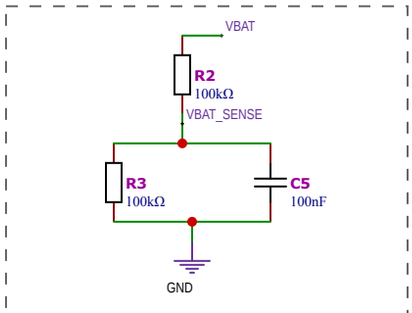
2.LDO VOLTAGE REGULATION(3.3V)



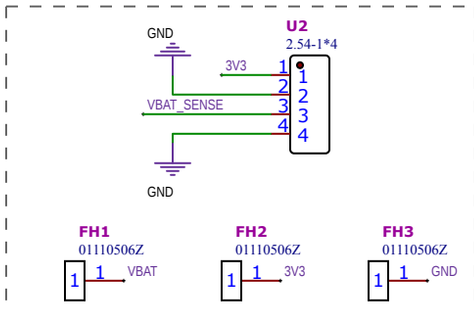
3.POWER STATUS LED



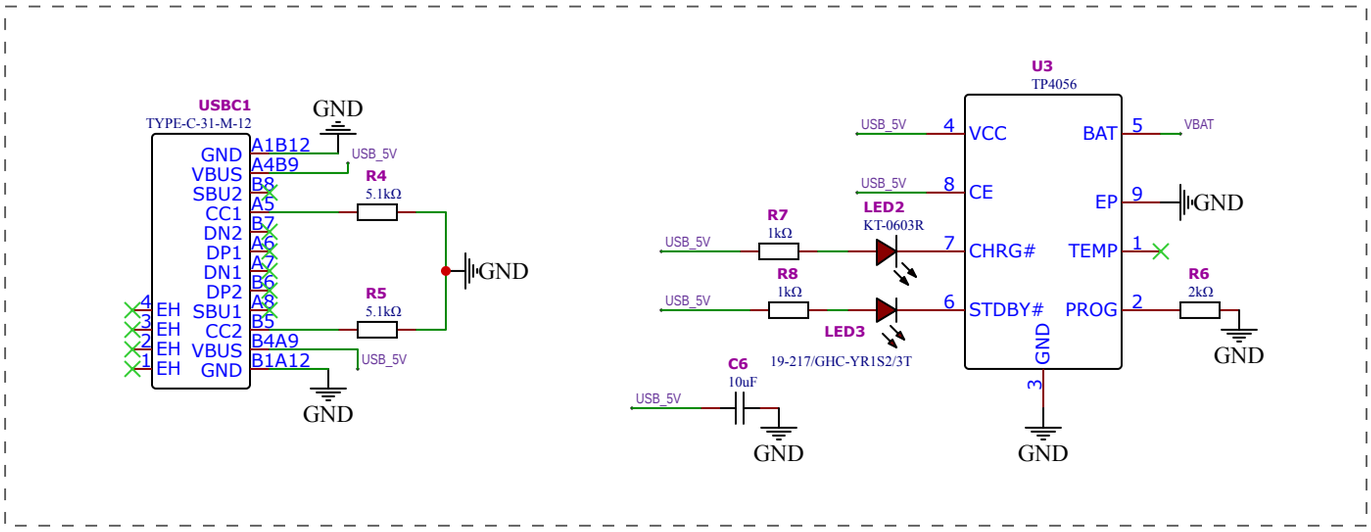
4.BATTERY VOLTAGE MONITOR (ADC DIVIDER)



6.ESP32 OUTPUT INTERFACE & TEST POINTS



5.USB-C Li-Ion CHARGING CIRCUI



TITLE: ESP32 Li-Ion Battery PSU(ass1 partA)		REV: 1.0
EasyEDA	Company: Heuronics pvt ltd.	Sheet: 1/1
	Date: 2026-05-29 Drawn By: Shubham Gupta	